

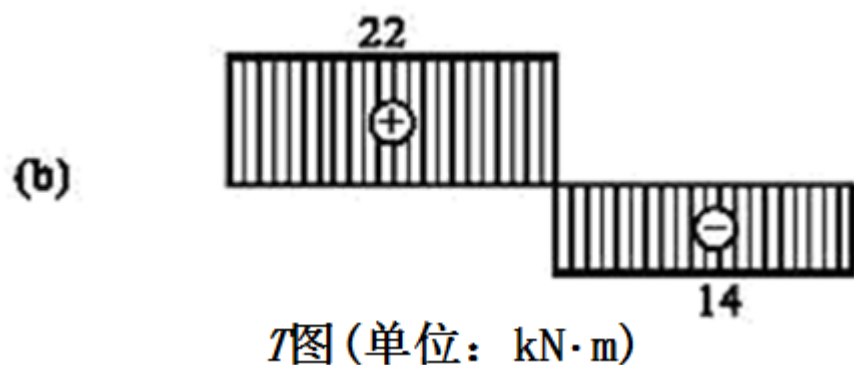
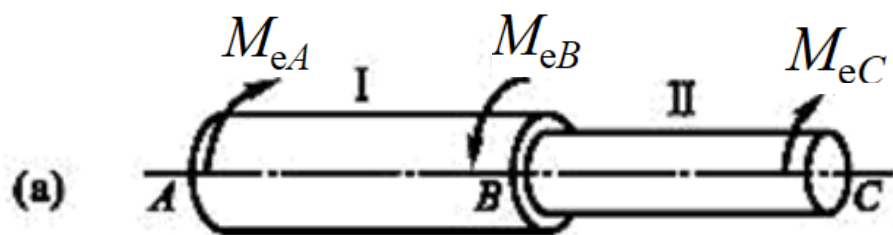


# 材料力学—习题课

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**例1：**图示阶梯状圆轴， $AB$ 段直径 $d_1 = 120 \text{ mm}$ ， $BC$ 段直径 $d_2 = 100 \text{ mm}$ 。扭转力偶矩 $M_{eA} = 22 \text{ kN}\cdot\text{m}$ ， $M_{eB} = 36 \text{ kN}\cdot\text{m}$ ， $M_{eC} = 14 \text{ kN}\cdot\text{m}$ ，材料的许用剪应力 $[\tau] = 80 \text{ MPa}$ 。试校核该轴的强度

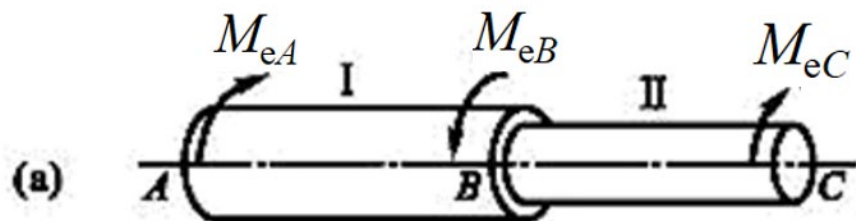


**解：(1) 绘扭矩图**

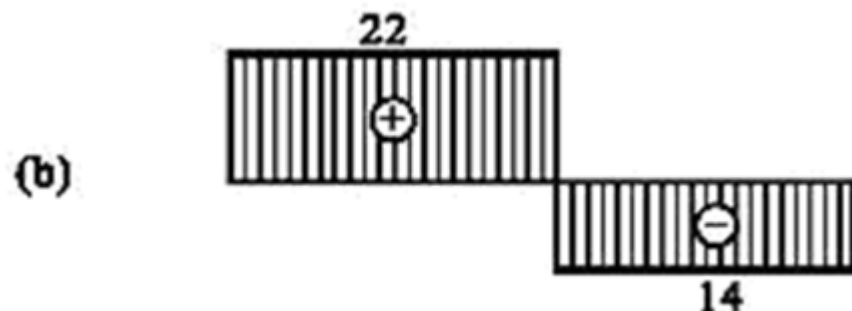
**(2) 求每段轴的横截面上的最大剪应力**

**$AB$ 段内**

$$\begin{aligned}\tau_{1,\max} &= \frac{T_1}{W_{p1}} = \frac{22 \times 10^3 \text{ N}\cdot\text{m}}{\frac{\pi}{16} (120 \times 10^{-3} \text{ m})^3} \\ &= 64.8 \times 10^6 \text{ Pa} = 64.8 \text{ MPa}\end{aligned}$$



**BC段内**



T图 (单位: kN·m)

$$\begin{aligned}\tau_{2,\max} &= \frac{T_2}{W_{p2}} = \frac{14 \times 10^3 \text{ N} \cdot \text{m}}{\frac{\pi}{16} (100 \times 10^{-3} \text{ m})^3} \\ &= 71.3 \times 10^6 \text{ Pa} = 71.3 \text{ MPa}\end{aligned}$$

### (3) 校核强度

$\tau_{2,\max} > \tau_{1,\max}$ , 但有  $\tau_{2,\max} < [\tau] = 80 \text{ MPa}$ , 故该轴满足强度条件

需要指出的是, 阶梯状圆轴在两段的连接处仍有**应力集中**现象, 在以上计算中对此并未考核

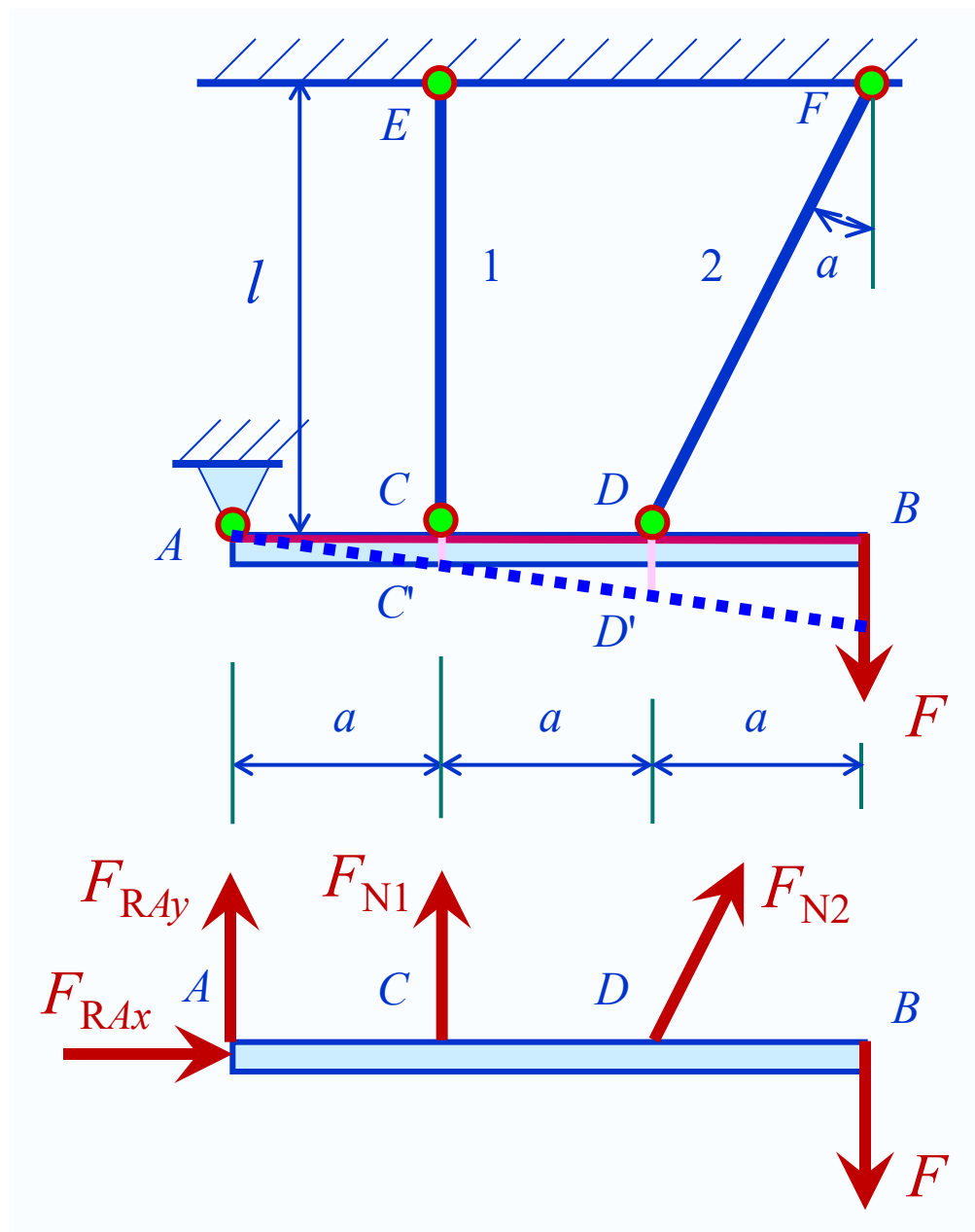
**例2：已知AB为刚性梁，  
1、2两杆的横截面面积  
相等，材料相同。求：  
1、2两杆的内力**

**解：超静定次数为1次，设变  
形后C、D点移至C'、D'点**

**(1) 静力平衡方程**

$$\sum M_A = 0$$

$$F_{N1}a + F_{N2} \cos \alpha \cdot 2a - F \cdot 3a = 0$$



## (2) 变形协调方程

$$\overline{DD'} = 2\Delta l_1 \quad \overline{DD'} = \frac{\Delta l_2}{\cos \alpha}$$

$$\frac{\Delta l_2}{\cos \alpha} = 2\Delta l_1$$

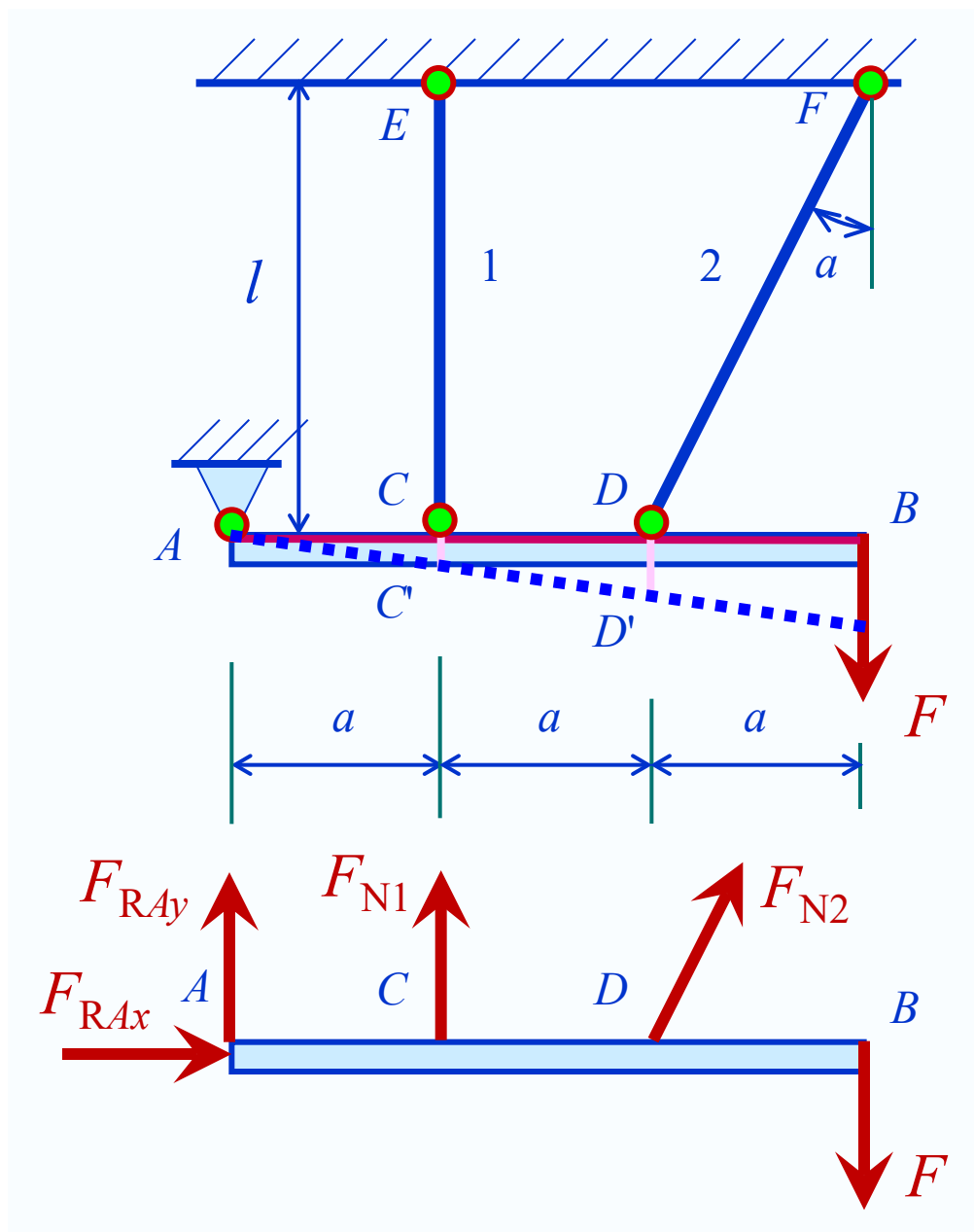
## (3) 物理方程

$$\Delta l_1 = \frac{F_{N1} l}{EA} \quad \Delta l_2 = \frac{F_{N2} l}{EA \cos \alpha}$$

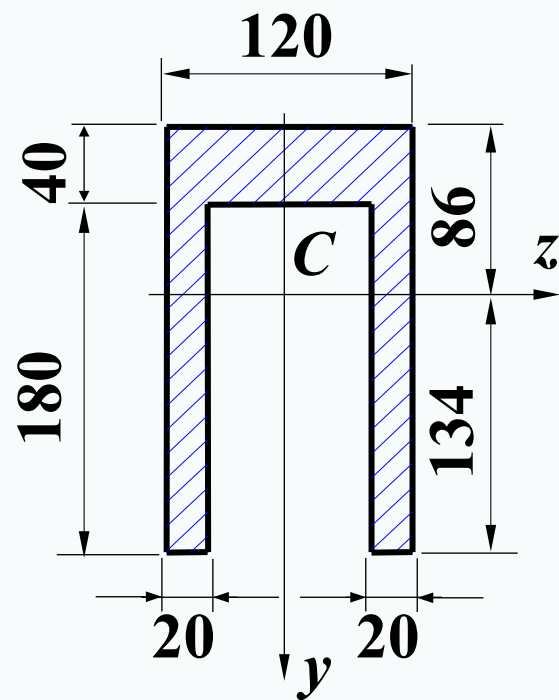
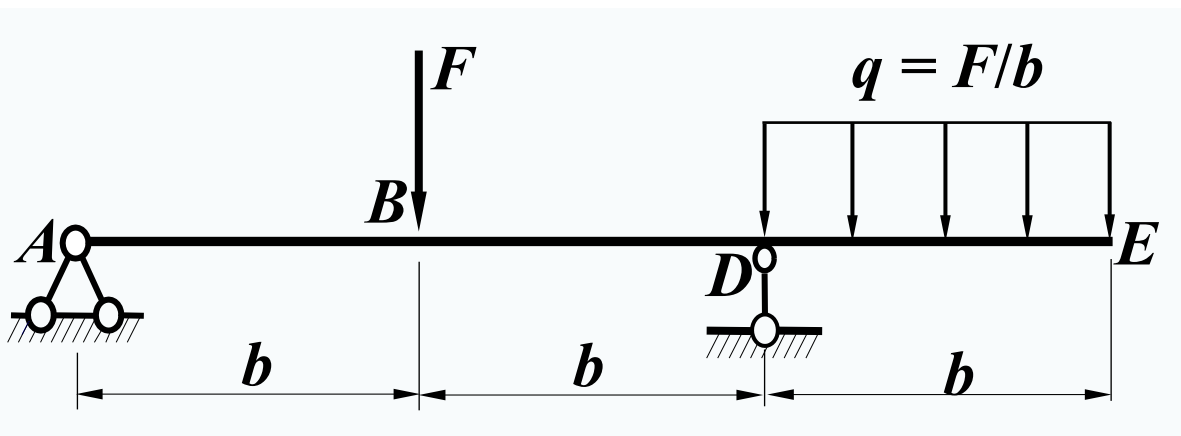
## (4) 联立求解

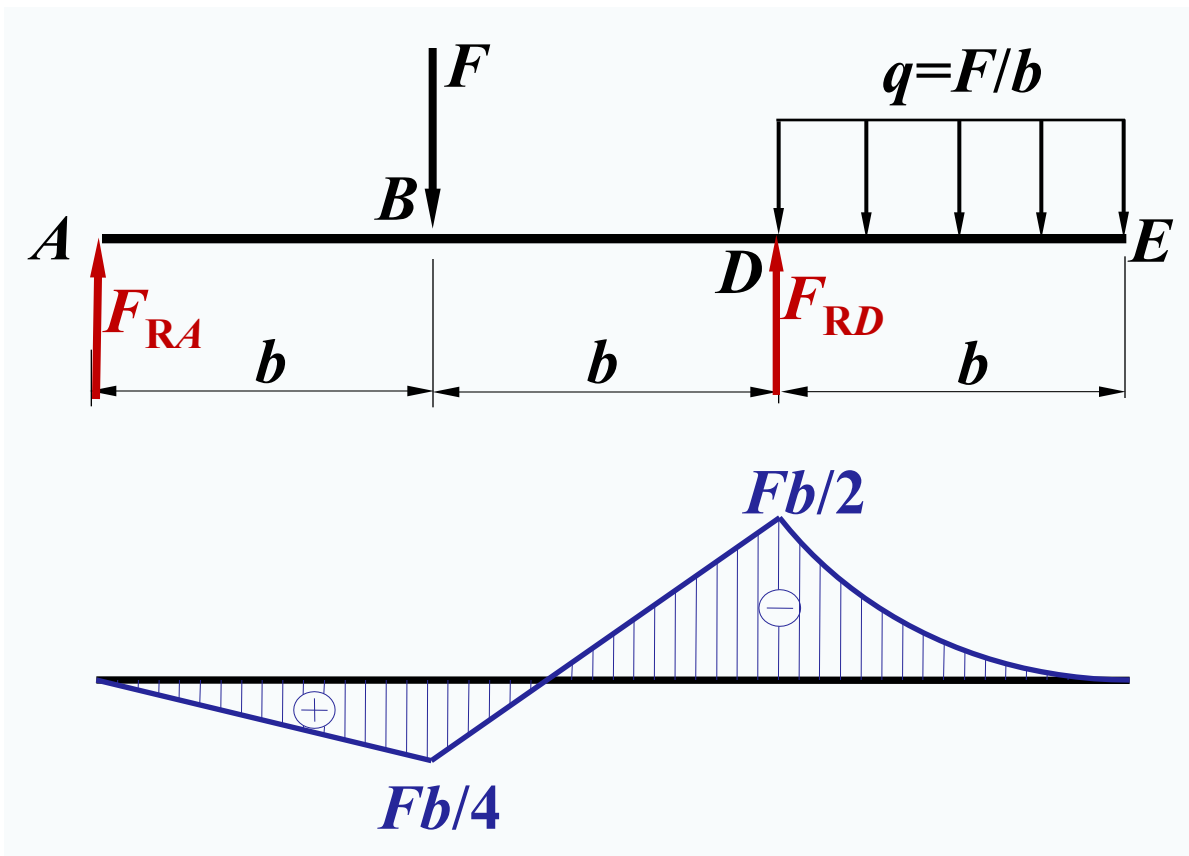
$$F_{N1} = \frac{3F}{4\cos^3 \alpha + 1}$$

$$F_{N2} = \frac{6F \cos^2 \alpha}{4\cos^3 \alpha + 1}$$



**例3：**图示槽形截面铸铁梁，已知： $b = 2 \text{ m}$ ，铸铁的许用拉应力  $[\sigma_t] = 30 \text{ MPa}$ ，许用压应力  $[\sigma_c] = 90 \text{ MPa}$ 。截面对中性轴的惯性矩  $I_z = 5493 \times 10^4 \text{ mm}^4$ ，作图示梁的剪力图和弯矩图，求梁的许可荷载  $[F]$



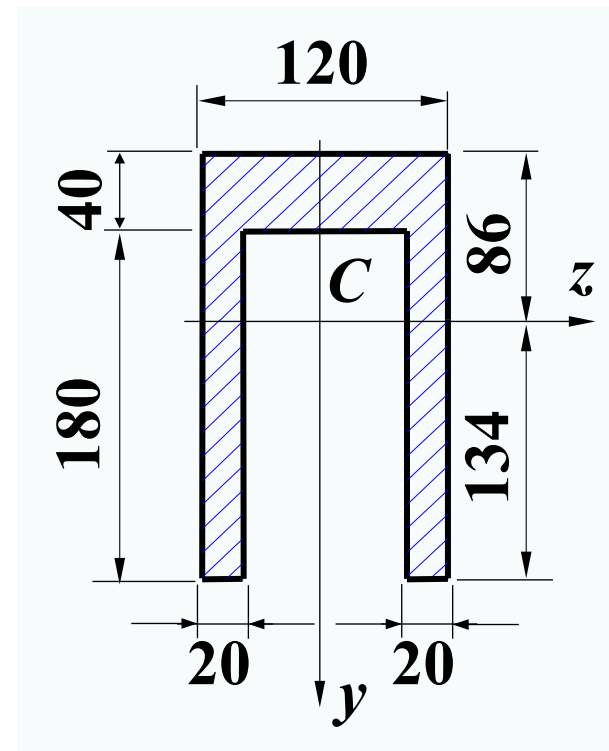
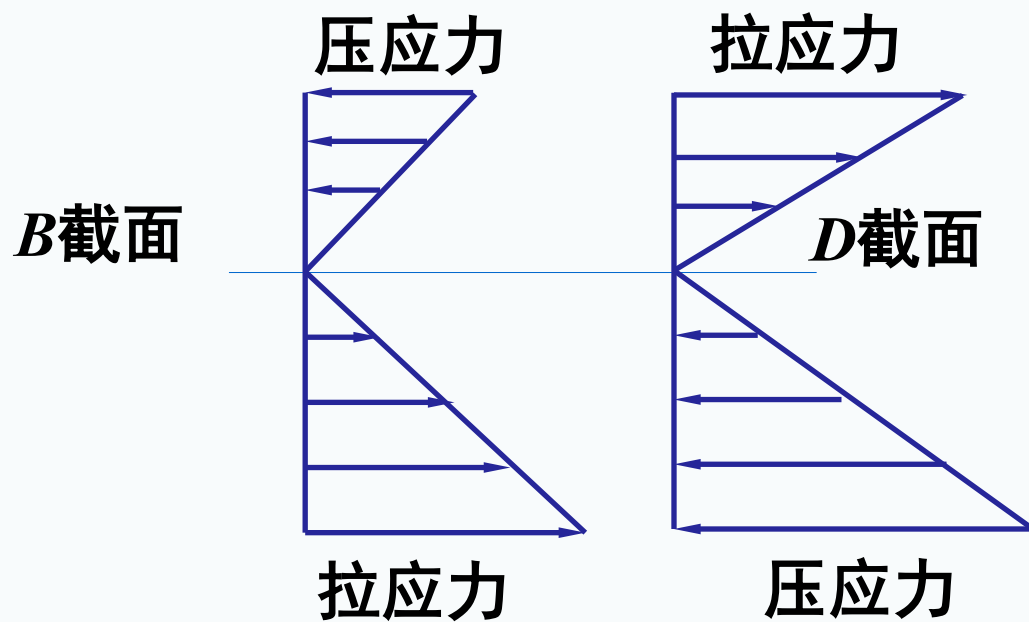
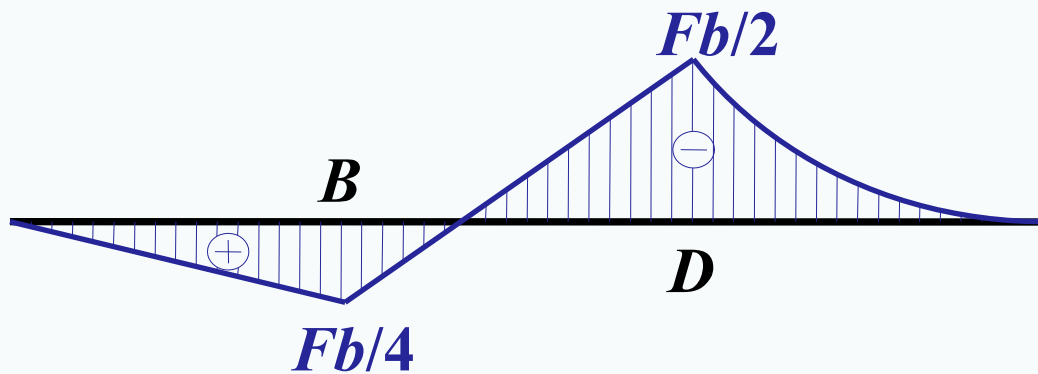


解：(1) 梁的约束力为

$$F_{RA} = \frac{F}{4} \quad F_{RD} = \frac{7F}{4}$$

$$M_{\max}^{+} = \frac{Fb}{4} \quad \text{发生在截面} B$$

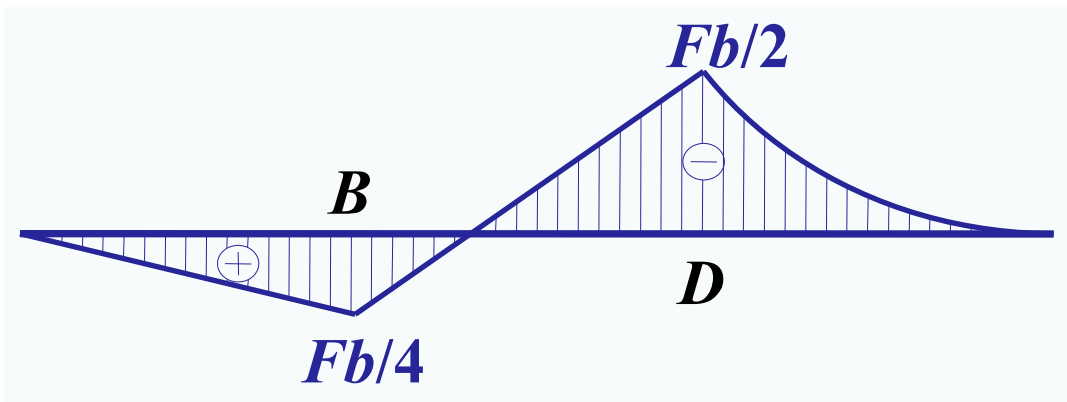
$$M_{\max}^{-} = \frac{Fb}{2} \quad \text{发生在截面} D$$



(2) 计算最大拉、压正应力

可见压应力强度条件由D截面控制，拉应力强度条件则B、D截面都要考虑



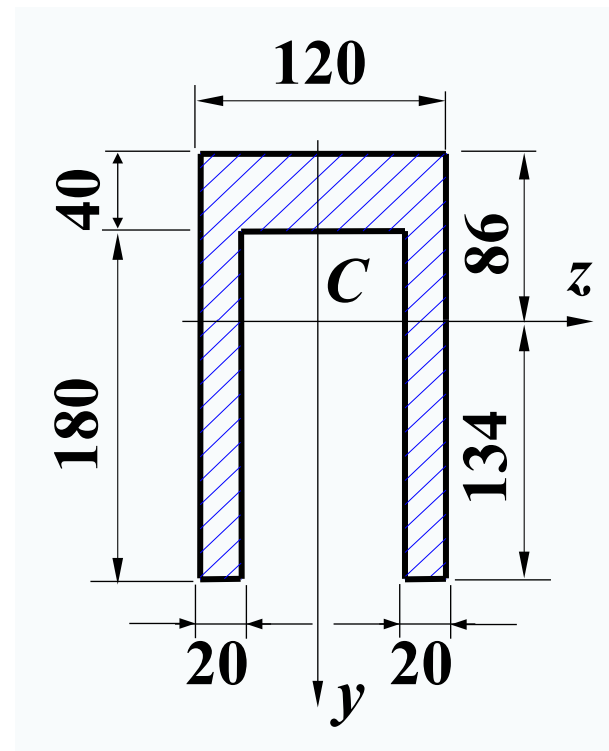


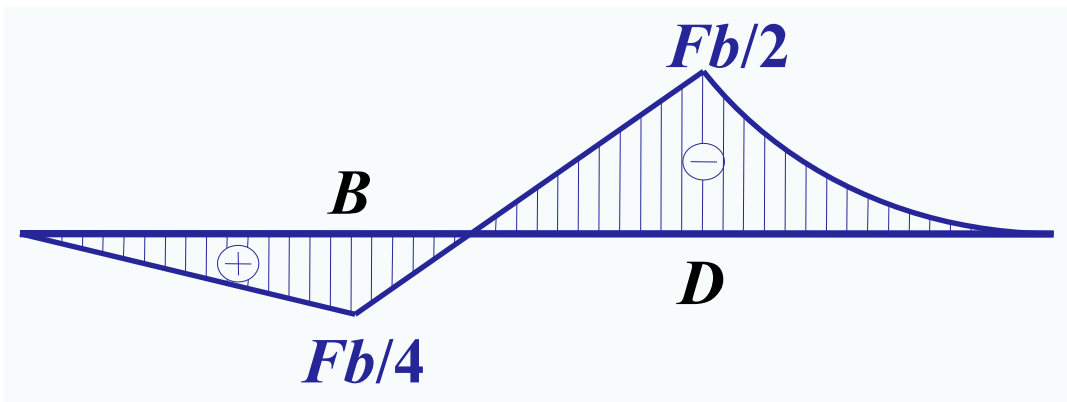
考虑截面B

$$\sigma_{t, \max} = \frac{M_B y_1}{I_z} = \frac{\left(\frac{F}{4} \times 2 \text{ m}\right) \times 0.134 \text{ m}}{5493 \times 10^{-8} \text{ m}^4} \leq 30 \text{ MPa} \quad F \leq 24.6 \text{ kN}$$

考虑截面D

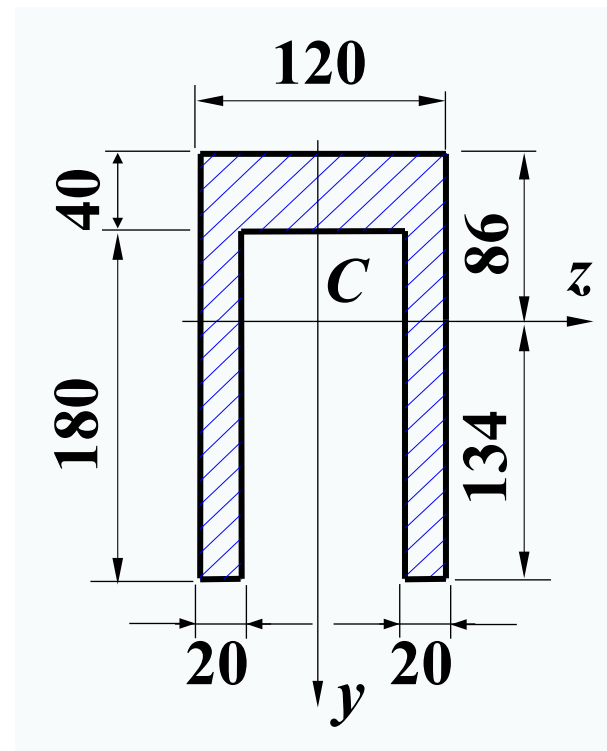
$$\sigma_{t, \max} = \frac{M_D y_2}{I_z} = \frac{\left(\frac{F}{2} \times 2 \text{ m}\right) \times 0.086 \text{ m}}{5493 \times 10^{-8} \text{ m}^4} \leq 30 \text{ MPa} \quad F \leq 19.2 \text{ kN}$$





考虑截面D

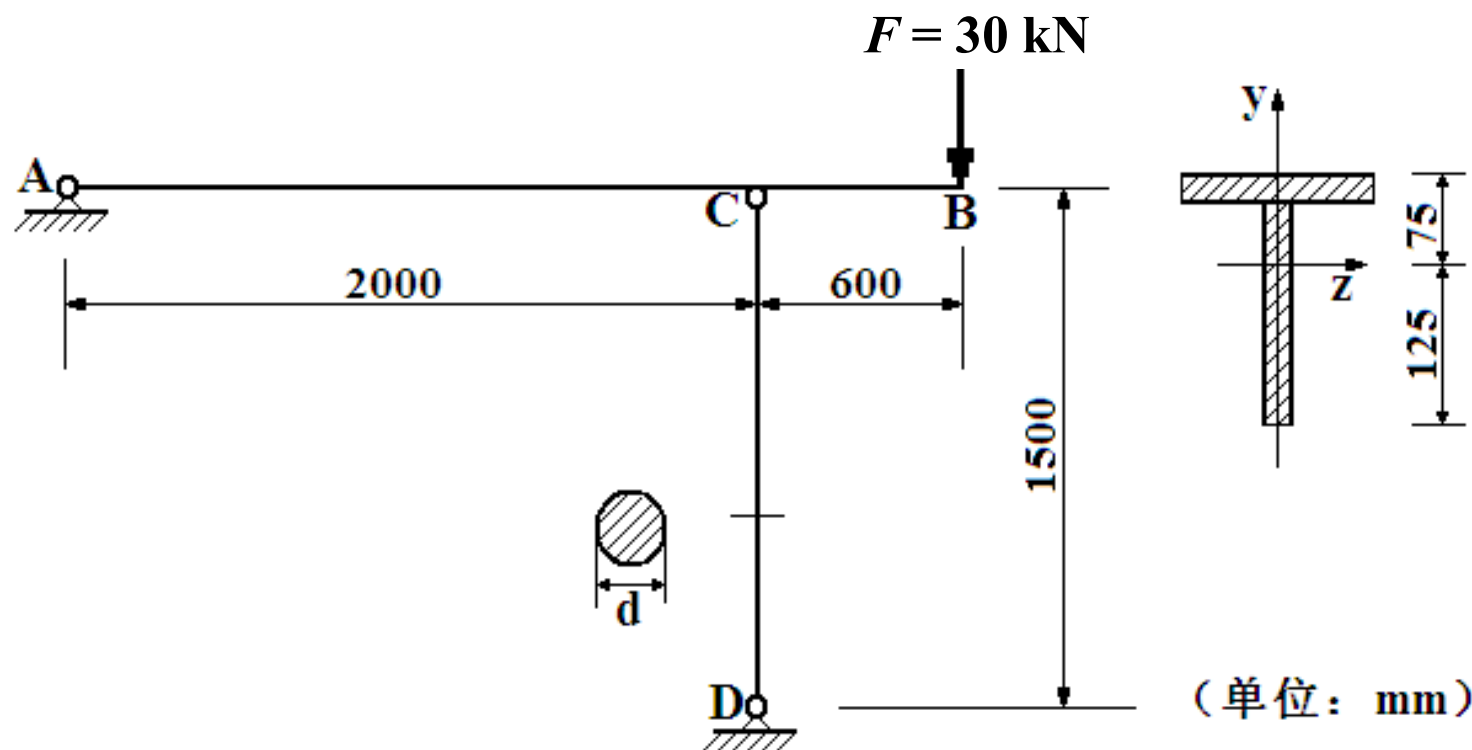
$$\sigma_{c, \max} = \frac{M_D y_1}{I_z} = \frac{\left( \frac{F}{2} \times 2 \text{ m} \right) \times 0.134 \text{ m}}{5493 \times 10^{-8} \text{ m}^4} \leq 90 \text{ MPa} \quad F \leq 73.8 \text{ kN}$$



因此梁的强度由截面D上的最大拉应力控制

$$[F] = 19.2 \text{ kN}$$

**例4：**如图所示，已知梁AB是T字形截面铸铁梁，惯性矩  $I_z = 5315 \text{ cm}^4$ ，许用拉应力  $[\sigma_t] = 50 \text{ MPa}$ ，许用压应力  $[\sigma_c] = 200 \text{ MPa}$ ，杆CD是两端铰支的圆截面压杆，直径为  $d = 40 \text{ mm}$ ，材料为Q235钢，弹性模量  $E = 200 \text{ GPa}$ ， $\sigma_p = 200 \text{ MPa}$ ，稳定安全因数  $n_{st} = 2.5$ 。试校核梁AB的强度和杆CD的稳定性

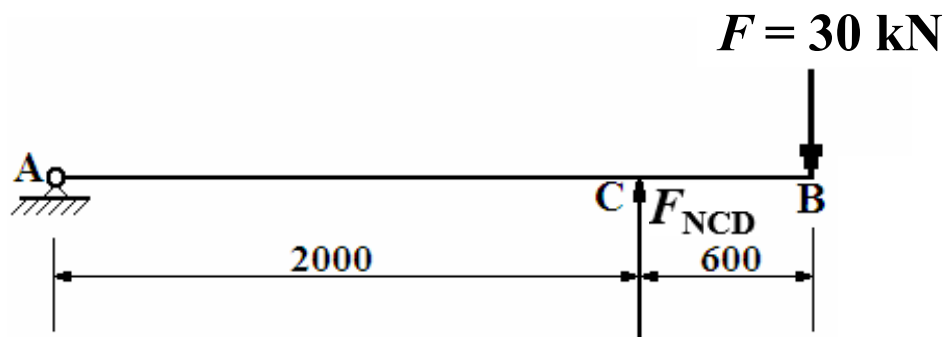


## 解：(1) 求CD杆的轴力

由  $\sum M_A = 0$

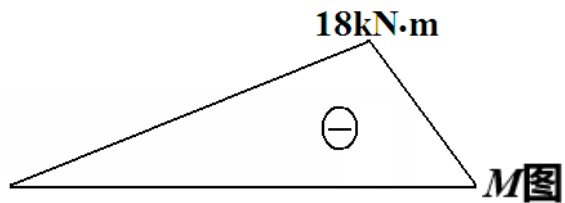
$$\therefore F_{\text{NCD}} \times 2000 - 30 \times 2600 = 0$$

$$\therefore F_{\text{NCD}} = 39 \text{ kN}$$



## (2) 校核AB梁的强度

作梁的弯矩图，由弯矩图可知，C截面为危险截面，上端受拉，下端受压



$$\sigma_{\text{t, max}}^{\text{C}} = \frac{M_{\text{C}} \cdot y_1}{I_z} = \frac{18 \times 10^3 \times 75 \times 10^{-3}}{5315 \times 10^{-8}} \times 10^{-6} = 25.4 \text{ MPa} < [\sigma_{\text{t}}]$$

$$\sigma_{\text{c, max}}^{\text{C}} = \frac{M_{\text{C}} \cdot y_2}{I_z} = \frac{18 \times 10^3 \times 125 \times 10^{-3}}{5315 \times 10^{-8}} \times 10^{-6} = 42.3 \text{ MPa} < [\sigma_{\text{c}}]$$

满足强度条件

### (3) 校核CD杆的稳定性

#### 判断CD杆的类型

$$\lambda = \frac{\mu l}{i} = \frac{1 \times 1.5 \times 10^3 \times 4}{40} = 150$$

$$\lambda > \lambda_p$$

$$\lambda_p = \sqrt{\frac{\pi^2 E}{\sigma_p}} = \sqrt{\frac{3.14^2 \times 200 \times 10^3}{200}} = 100$$

所以CD杆为细长杆

$$F_{cr} = \frac{\pi^2 EI}{(\mu l)^2} = \frac{3.14^2 \times 200 \times 10^9 \times 3.14 \times 40^4 \times 10^{-12}}{64 \times 1.5^2} \times 10^{-3} = 110 \text{ kN}$$

$$[F] = \frac{F_{cr}}{n_{st}} = \frac{110}{2.5} = 44 \text{ kN} \quad F_{NCD} < [F]$$

所以CD杆满足稳定性条件

**例5：**直径为 $d = 0.1 \text{ m}$ 的圆杆受力如图， $M = 5 \text{ kN}\cdot\text{m}$ ， $F = 100 \text{ kN}$ ，为铸铁构件， $[\sigma_t] = 40 \text{ MPa}$ ， $[\sigma_c] = 150 \text{ MPa}$ ，试用强度理论校核杆的强度

**解：**危险点 $A$ 的应力状态如图：

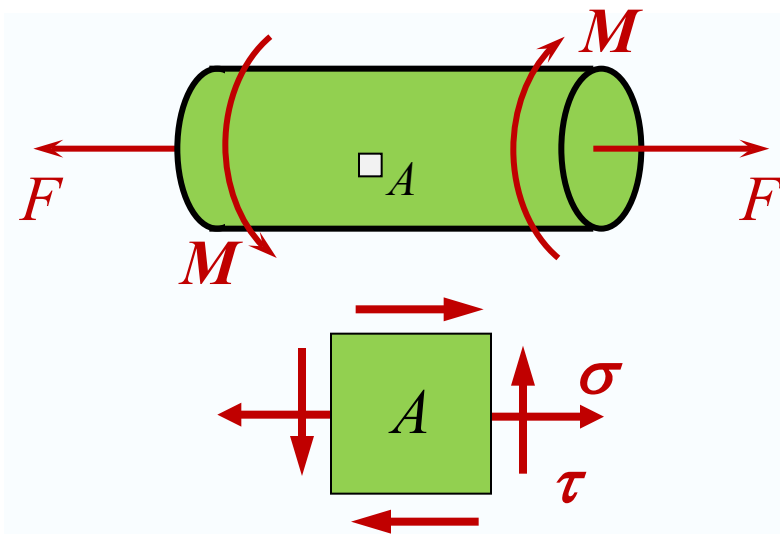
$$\sigma = \frac{F}{A} = \frac{4 \times 100}{\pi \times 100^2} \times 10^3 = 12.73 \text{ MPa}$$

$$\tau = \frac{T}{W_p} = \frac{16 \times 5}{\pi \times 100^3} \times 10^6 = 25.46 \text{ MPa}$$

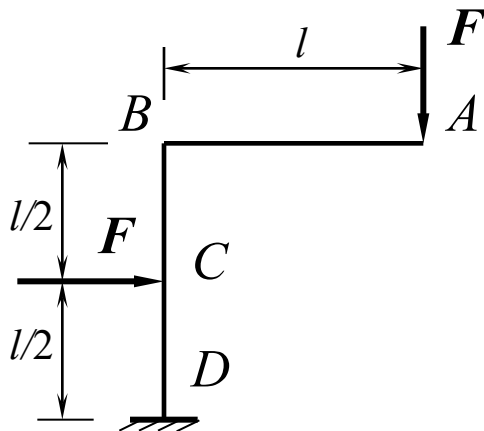
$$\left. \begin{matrix} \sigma'_1 \\ \sigma'_2 \end{matrix} \right\} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} = \frac{12.73}{2} \pm \sqrt{\left(\frac{12.73}{2}\right)^2 + 25.46^2} = \begin{cases} 32.61 \\ -19.88 \end{cases} \text{ MPa}$$

$$\therefore \sigma_1 = 32.61 \text{ MPa} \quad \sigma_2 = 0, \sigma_3 = -19.88 \text{ MPa}$$

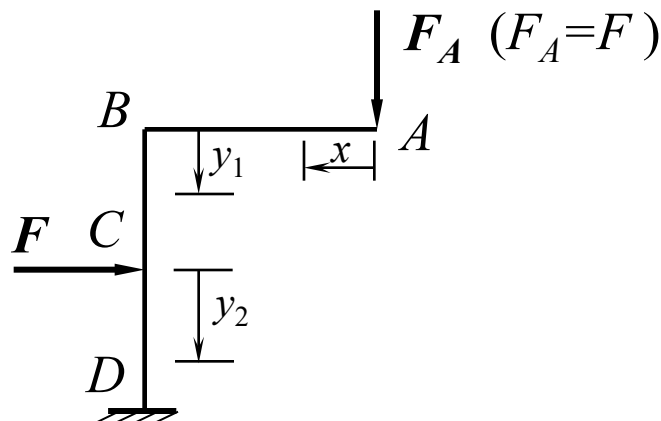
**脆性材料，以拉为主**  $\sigma_{r1} = \sigma_1 < [\sigma]$  **故，安全**



**例5：**图示刚架各杆的弯曲刚度均为 $EI$ ，不计剪力和轴力对位移的影响。试用卡氏第二定理求 $A$ 截面的铅垂位移 $\Delta_{Ay}$



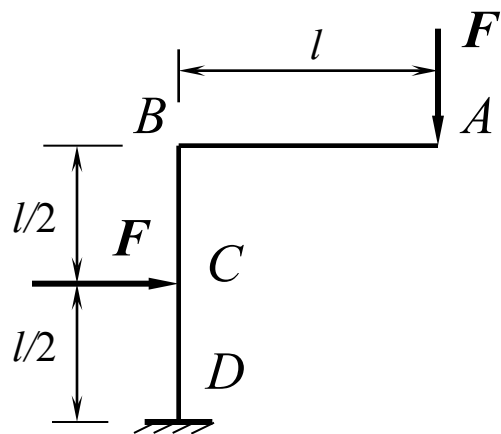
(a)



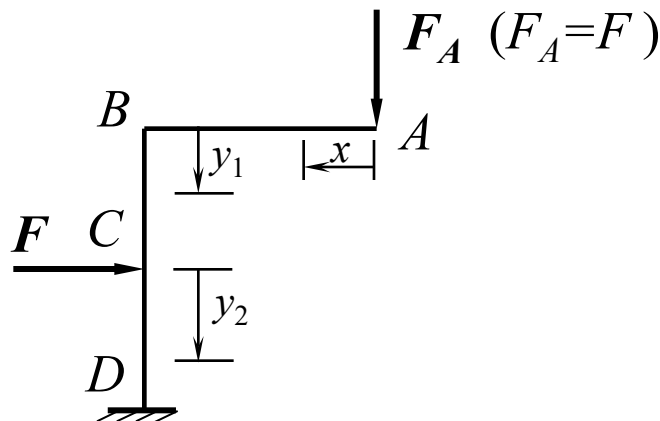
(b)

**解：**由于刚架上 $A$ 和 $C$ 截面的外力均为 $F$ ，求 $A$ 截面的铅垂位移时，应将 $A$ 处的力 $F$ 和 $C$ 处的力 $F$ 区别开(图b)，在应用卡氏第二定理后，令 $F_A = F$ ，即

$$\Delta_{Ay} = \left. \frac{\partial V_{\varepsilon}}{\partial F_A} \right|_{F_A = F}$$



(a)



(b)

各段的弯矩方程及其对 $F_A$ 的偏导数分别为

$$AB\text{段 } (0 \leq x \leq l) \quad M(x) = -F_A x \quad \frac{\partial M(x)}{\partial F_A} = -x$$

$$BC\text{段 } (0 \leq y_1 \leq l/2) \quad M(y_1) = -F_A l \quad \frac{\partial M(y_1)}{\partial F_A} = -l$$



**CD段** ( $0 \leq y_2 \leq l/2$ )  $M(y_2) = -F_A l - F y_2$   $\frac{\partial M(y_2)}{\partial F_A} = -l$

**由卡氏第二定理得**

$$\Delta_{Ay} = \left. \frac{\partial V_{\varepsilon}}{\partial F_A} \right|_{F_A = F}$$

$$\Delta_{Ay} = \frac{1}{EI} \left[ \int_0^l F x^2 \, dx + \int_0^{l/2} F l^2 \, dy_1 + \int_0^{l/2} (F l^2 + F l y_2) \, dy_2 \right]$$

$$= \frac{35 F l^3}{24 EI} \quad (\downarrow) \quad (\text{已令各弯矩方程中的 } F_A = F)$$